

Title: Contrastive Representation Learning for RGB–Depth Fusion in Visual Perception

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Project Description

Learning effective representations from multiple visual modalities—such as RGB and Depth—is a key challenge in computer vision, especially in environments where each modality provides unique and complementary information. Traditional fusion methods often rely on supervised training and heuristic feature combinations, which can limit generalization and scalability. This project is motivated by the need to explore alternative approaches that can align modalities using less supervision while still preserving rich semantic structure. Specifically, it investigates how contrastive learning can be used to learn shared embeddings for RGB and Depth that are both discriminative and modality-invariant.

This project proposes to explore **contrastive representation learning** as a framework to align RGB and Depth modalities in a shared embedding space, encouraging cross-modal consistency without the need for extensive labeled data. Inspired by methods such as SimCLR [1], and recent cross-modal extensions like CrossCLR [2] and pixel-level contrastive RGB-D fusion [3], this approach aims to fuse visual modalities at the representation level, rather than relying solely on early or mid-level fusion heuristics.

While contrastive learning typically benefits from large datasets, this project leverages pretrained visual encoders and trains only projection heads, reducing both computational and data requirements. A moderate-sized RGB-D dataset will be used to evaluate whether contrastive alignment improves representation quality for downstream tasks like scene classification or semantic segmentation. The project directly supports the development of modality-agnostic perceptual models, a key objective of ongoing efforts in foundation model research and multimodal sensing.

Data Set

This project will use the NYU Depth v2 (NYUv2) dataset [4], a widely used benchmark in RGB-D computer vision. Collected using a Microsoft Kinect sensor, NYUv2 contains over 1,400 labeled indoor scene frames captured across a variety of environments such as offices, bedrooms, and kitchens. Each frame includes a high-resolution RGB image, a spatially aligned depth map, a label for the scene type of the capture and dense semantic segmentation labels for over 800 object categories. Its moderate size and rich annotations make it well-suited for prototyping contrastive representation learning and evaluating aligned embeddings on downstream tasks such as scene classification and segmentation.

References

- [1] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, “A simple framework for contrastive learning of visual representations,” in *ICML*, 2020.
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 - [3] J. Wu, F. Hao, W. Liang, and J. Xu, “Transformer fusion and pixel-level contrastive learning for RGB-D salient object detection,” *IEEE Transactions on Multimedia*, 2023.
 - [4] N. Silberman, D. Hoiem, P. Kohli, and R. Fergus, “Indoor segmentation and support inference from RGBD images,” in *Proceedings of the European Conference on Computer Vision (ECCV)*, pp. 746–760, Springer, 2012.
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